

1. Administrative & Study Identification

Full Study Title: A Phase 3, Randomized, Double-Blind, Placebo-Controlled, 3-Arm, 3-Period Study to Assess the Efficacy and Safety of a New Formulation of Oral Cladribine Compared With Placebo in Participants With Generalized Myasthenia Gravis (MyClad) **Short Title/Acronym:** MyClad **Posting ID:** 276832736207126425 **Protocol Number:** MS700568_0183 **NCT Number:** NCT06463587 **Trial Status:** RECRUITING **Sponsor/Company Name:** Merck (Merck Healthcare KGaA, Darmstadt, Germany / EMD Serono) **Investigational Product:** Oral Cladribine Formulation (Drug Name: cladribine | Trade Name: Mavenclad | ATC Codes: L01BB04, L04AA40) **Phase of Development:** Phase III **Medical Monitor:** eMediUSA@emdserono.com / 888-275-7376

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine the efficacy and safety of a new oral cladribine formulation in participants with Generalized Myasthenia Gravis (gMG) in comparison to placebo. **Secondary Objectives:** To investigate the sustained efficacy, the need for retreatment, and the long-term safety of oral cladribine in gMG, as well as to characterize the Pharmacokinetics (PK) of the new formulation.

2.2 Background & Rationale Generalized Myasthenia Gravis (gMG) is a rare, chronic autoimmune neuromuscular disorder characterized by debilitating muscle weakness and fatigue. Cladribine selectively targets the B and T lymphocytes believed to contribute to the production of harmful autoantibodies that cause inflammation and damage to the neuromuscular junction. If approved, cladribine capsules would be the first oral treatment for gMG, potentially reducing the burden of the disease by addressing its underlying cause through a short-course, infrequently-administered oral regimen taken at home.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled (3-Arm: Cladribine Low Dose, Cladribine High Dose, Placebo) **Blinding:** Quadruple-blind (Participant, Care Provider, Investigator, Outcomes Assessor) **Randomization:** Randomized (Sequential Assignment)

3.2 Planned Enrollment & Duration Target **\$N\$:** 264 participants
Number of Sites: Global / Multicenter **Estimated Study Start:** 25-Jun-2024 **Estimated Primary Completion:** 19-May-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years old at the time of signing the informed consent. 2. Diagnosis of Myasthenia Gravis with generalized muscle weakness, meeting clinical criteria for Myasthenia Gravis Foundation of America (MGFA) Class II to IVa classification. 3. Tested positive for Acetylcholine receptor antibody (anti-AChR) or muscle-specific kinase antibody (anti-MuSK); OR autoantibody seronegative but positive for anti-low-density lipoprotein receptor-related protein 4 antibodies (anti-LRP4). 4. Screening and Baseline MG-ADL score ≥ 6 with $\geq 50\%$ of the total score due to non-ocular symptoms, and scores must be stable (difference ≤ 2). 5. Body weight ≥ 40 kg. 6. If on oral corticosteroids or acetylcholinesterase inhibitors, must be on stable daily doses for at least 3 months prior to and during screening.

4.2 Exclusion Criteria 1. Unstable condition on current gMG background medication or history of myasthenic crisis in the last 12 months prior to and during screening. 2. Recent use of exclusionary immunosuppressive/immunomodulatory therapies (e.g., azathioprine, mycophenolate mofetil, methotrexate, cyclosporine, tacrolimus within 4 weeks prior to randomization; eculizumab, rozanolixizumab, efgartigimod, ravulizumab, or zilucoplan within 8 weeks prior). 3. Active, clinically significant infection, active malignancy, history of tuberculosis (TB), or negative for Varicella Zoster Virus antibodies at screening. 4. History of thymectomy within 6 months prior to Screening.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants are randomized to receive two short courses of cladribine (low-dose or high-dose) or matched oral placebo. The courses are separated by 4 weeks. **Route of Administration:** Oral (Capsules) **Duration of Treatment:** The study spans approximately 3 years, divided into 3 periods: the double-blind placebo control (DBPC) pivotal period, the blinded extension (BE), and the retreatment (RT) period.

5.2 Concomitant & Prohibited Therapy Permitted: Stable doses of baseline standard-of-care medications for gMG. **Prohibited:** Use of other concurrent investigational drugs or specific prohibited immunosuppressants/biologics as defined in the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window	Key Activities/Procedures
Screening	Up to Day 0	Informed Consent, Medical History, Antibody Testing, Baseline MG-ADL, QMG, MGC
Periods 1 & 2	Weeks 0 to End of BE	Onsite visits every 4 weeks with telehealth calls between visits; IP Administration (Courses 1 & 2)
Period 3 (RT)	Up to Year 3	Onsite visits every 12 weeks, with two telehealth calls in between each site visit; Retreatment assessment
End of Study	Year 3	Final Safety Labs, Efficacy Assessment, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change from Baseline in Myasthenia Gravis - Activities of Daily Living (MG-ADL) Scale Score at Week 24 During the Double-Blind Placebo Controlled (DBPC) Period. **Measurement Method:** Clinical assessment questionnaire evaluated by an outcomes assessor.

7.2 Secondary & Exploratory Endpoints

- * **Secondary Endpoint 1:** Change from Baseline in Quantitative Myasthenia Gravis (QMG) Scale Score at Week 24 During the Double-Blind Placebo Controlled (DBPC) Period.
- * **Secondary Endpoint 2:** Percentage of MG-ADL Responders at Week 24 During the Double-Blind Placebo Controlled (DBPC) Period.
- * **Secondary Endpoint 3:** Change from Baseline in Myasthenia Gravis Composite (MGC) Scale Score at Week 24 During the Double-Blind Placebo Controlled (DBPC) Period.
- * Sustained efficacy, duration of treatment effects, and need for retreatment over the 3-year period.
- * Long-term safety, tolerability, and Pharmacokinetics (PK).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via regular onsite visits (every 4 to 12 weeks) and frequent telehealth follow-up calls between site visits. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee to oversee the safety of the 3-arm Phase 3 trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for AE evaluations. **Sample Size Justification:** \$N = 264\$ participants, statistically powered to detect a meaningful change in MG-ADL scores at Week 24 between the active arms and placebo. **Handling of Missing Data:** Standard handling methods

(e.g., Mixed-Model Repeated Measures [MMRM] or Multiple Imputation) for longitudinal MG-ADL, QMG, and MGC scores.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: High-touch monitoring model utilizing both onsite visits and telehealth calls to ensure protocol adherence.

Audit Trail: Validated eCRF locking and data clarification forms.

11. Study Identifiers & Governance

Primary ID: MS700568_0183 **Posting ID:** 276832736207126425 **Registry Number:** NCT06463587 **Sponsor/Lead Organization:** Merck **Study Title:** MyClad **Official Regulatory Title:** A Phase 3, Randomized, Double-Blind, Placebo-Controlled, 3-Arm, 3-Period Study to Assess the Efficacy and Safety of a New Formulation of Oral Cladribine Compared With Placebo in Participants With Generalized Myasthenia Gravis (MyClad)

12. Clinical Framework (Generalized Myasthenia Gravis Specific)

Primary Condition / Disease Area: Generalized Myasthenia Gravis (gMG) **Preferred Name:** Myasthenia Gravis, Generalized **Preferred UMLS Name:** Generalized Myasthenia Gravis **MeSH Code:** D009157

SNOMED ID: 722510008 **Diagnosis Threshold:** Presence of anti-AChR, anti-MuSK, or anti-LRP4 autoantibodies with moderate to severe muscle weakness (MGFA Class II-IVa) **Medical Methodology:** Selective targeting of B and T lymphocytes contributing to autoantibody production **Optimization Target:** Significant reduction in MG-ADL, QMG, and MGC scores and long-term symptom control

13. Study Procedures & Methodology

Management Pathway: Oral administration of disease-modifying therapy **Education Protocol (HCP):** Standardized training for administering the MG-ADL, QMG, MGC, and other clinical

assessments **Education Protocol (Patient):** Strict guidance on the short-course oral regimen taken at home and telehealth compliance

Quality Audit Mechanism: Tracking of adherence through rigorous scheduled telehealth check-ins and site visits

14. Observation & Follow-Up Schedule

Total Duration: Approximately 3 years **Onsite Visit Cadence:** Every 4 weeks (Periods 1 and 2), transitioning to every 12 weeks (Period 3) **Remote Monitoring Frequency:** 1 to 2 telehealth calls scheduled between every onsite visit **Checklist Review Interval:** Routine review during all 50 planned touchpoints (27 in Periods 1 & 2; 23 in Period 3)

15. Sign-off & Approval

Role	Name	Signature	Date						
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]						
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]						
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]						